

# YIFAN HU

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## EDUCATION

### University of Pennsylvania

*M.S.E. in Data Science and Artificial Intelligence, GPA 3.94 / 4.0*

Coursework: Applied Machine Learning, Cloud & Internet Systems, Big Data Analytics, Database Systems

Philadelphia, PA

Sept 2025 – Expected May 2027

### New York University

*B.S. in Mathematics, Minor in Computer Science, Summa Cum Laude, GPA 3.96 / 4.0*

Honors: George Bachman Award in Mathematics (2025); Dean's List (2023–2024)

New York, NY

Sept 2023 – May 2025

## EXPERIENCE

### University of Pennsylvania, GMLR Lab

*Student Researcher, Generative Multimodal Learning & Reasoning Lab*

- Research directions: world models for robot control, and flow / diffusion-based robot policies; advised by Prof. Jiatao Gu.

Philadelphia, PA

Sept 2026 – Present

### ROBOTERA

*Robot Learning Intern, Embodied AI Team*

Beijing, China

May 2026 – Aug 2026

#### Tactile Sim-to-Real RL for Dexterous In-Hand Rotation | Live Demo at WAIC 2026 | [Video](#)

- Demonstrated **15+ min** of continuous, drop-free 360° rotation via **zero-shot sim-to-real** transfer on a 21-DoF XHand1 Pro; deployed a vision-free policy at 20 Hz on CPU, supporting cylinders 32–48 mm in diameter.
- Trained privileged PPO teachers in Isaac Lab on a 4-GPU cloud cluster, with 20,480 parallel environments per run and an independent seed on each GPU; randomized object properties, hand orientation, actuator gains and sensor noise.
- Adapted HORA/RMA-style teacher-student distillation: trained a temporal-convolution module on student rollouts to infer the teacher latent from 30-frame proprioceptive and tactile histories, removing privileged inputs at deployment.
- Aligned SDK control modes, tactile observations and warm-start grasping between simulation and hardware; built batch evaluations of drop rate, rotation speed and action smoothness under friction perturbations.

#### DexGen-Teleop: Manus Glove-Guided Diffusion for Dexterous Manipulation | [Video](#)

- Built Manus glove-guided teleoperation for a 21-DoF hand: the glove gives only high-level intent (direction, pace, stop) while the diffusion policy generates all finger motion; tuned the pipeline in MuJoCo before hardware deployment, demonstrating screwdriver and bottle rotation and integration with the ROBOTERA M7 humanoid.
- Collected **446K transitions** from noisy RL experts and trained a FiLM-conditioned UNet diffusion prior over fingertip motion plus an inverse-dynamics model; implemented DDIM gradient guidance with trust-region clamping.
- Designed a cycle-aware gesture decoder with direction commitment and stop detection; improved steady-state direction accuracy from **51% to 96%** and reduced spurious direction flips from 157 to 4.2/min in offline glove-session replay.
- Reduced CPU thread contention to achieve **9.8 ms** controller inference on 4 threads within a 20 Hz control loop; added hand-opening command filtering, pre-grasp initialization and verified control handover.

### Broadsense

*Algorithm Engineer Intern, Robotics & Perception*

Changsha, China

Jun 2025 – Aug 2025

- Built a ROS 2 road-slope estimator using only a 6-axis IMU, with a Kalman filter that estimates gyro bias online, adaptive noise under acceleration and turning, and quaternion IMU-to-vehicle alignment.
- Built a port-crane collision monitor on one RGB camera instead of stereo or LiDAR: a three-thread Depth Anything V2 pipeline on a dedicated CUDA stream, alarming on per-zone disparity and pixel-density thresholds to suppress noise.

## PROJECTS

### Notgoogle — Distributed Search Engine, UPenn CIS 5550 (Team Lead)

Spring 2026

- Led a 4-person team building a distributed Java search engine with a replicated KV store, MapReduce and PageRank on 5 AWS EC2 nodes; crawled **1M+ pages** at 2,000+ pages/min, approximately 3× baseline throughput.

### News Source Classification — DistilBERT Fine-tuning, UPenn CIS 5190 (Team Lead)

Fall 2025

- Expanded the labeled dataset from 3,815 to 91,216 URLs and fine-tuned DistilBERT to **92.9% test accuracy**, versus 66% for TF-IDF logistic regression and 79% for linear SVM; released the model on Hugging Face.

## SKILLS

**Robot Learning:** Isaac Sim / Isaac Lab, PPO, teacher-student distillation, domain randomization, diffusion policies, tactile sensing, MuJoCo, ROS 2, EtherCAT

**Programming:** Python, C++, Java, SQL, JavaScript / TypeScript

**Frameworks & Tools:** PyTorch, Transformers, scikit-learn, Docker, AWS, Git, Linux